

The formulae $\frac{\partial \rho U_i}{\partial t} + \frac{\partial}{\partial x_j} (\rho U_i U_j) = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial U_i}{\partial x_j} \right) + g_i (\rho - \rho_s)$ $\frac{\partial \rho U_i k}{\partial t} = \frac{\partial}{\partial x_j} \left(\left(\mu + \frac{\mu_T}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right) + P + G - \rho \epsilon$ for building $\frac{\partial \rho U_i \epsilon}{\partial t} = \frac{\partial}{\partial x_j} \left(\left(\mu + \frac{\mu_T}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right) + (C_1 - C_{1RHO}) \frac{\epsilon}{k} (P + C_2 G) - C_2 \rho \frac{\epsilon^2}{k}$ $\frac{\partial}{\partial x_i} (\rho U_i H) = \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial T}{\partial x_i} - \rho u'_i h \right)$ $-\rho u'_i u'_j = \mu_T \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta$ state of the art $\frac{\partial}{\partial x_j} (\rho U_i U_j) = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial U_i}{\partial x_j} - \rho u'_i u'_j \right) + g_i (\rho - \rho_s)$ $\frac{\partial \rho U_i \epsilon}{\partial t} = \frac{\partial}{\partial x_j} \left(\left(\mu + \frac{\mu_T}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right) + C_1 \frac{\epsilon}{k} (P + C_2 G) - C_2 \rho \frac{\epsilon^2}{k}$ biomedical research facilities.

NIH Central Utility Plant SOP

STANDARD OPERATING PROCEDURE (SOP) TO SOUND AND FILL MAIN FUEL OIL TANKS

Instructional Document





STANDARD OPERATING PROCEDURE	
Title: SOP to Sound and Fill Main Fuel Oil Tanks	
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I. INTRODUCTION

I.1 INTRODUCTION

This document reviews the procedure for operators to sound the main fuel oil tanks and the procedure to fill the fuel oil tanks.

I.2 SOUNDING MAIN FUEL TANKS

What is Sounding? – A pipe which leads down to almost the bottom of a tank to enable the depth of liquid to be measured by a sounding stick or tape.

- 1) Locate which tank you plan to sound (tank 1 or tank 2)



- 2) Open the Bilco door exposing the sounding tube.



- 3) Lower sounding stick into the sounding tube until it stops at the bottom of the tank.
- 4) Lift the sounding stick until you see fuel oil on the stick. (Chalk can be applied to the stick to make the fuel level visible if needed)



- 5) Record the measurement shown (i.e.: 125.75")
- 6) Use the Main Fuel Tank Capacity Chart located at Bldg. 58 and the log desk to convert the tank level to gallons. (Also located in this SOP in section 2 pg. 8)



7) Convert the measurement from inches to feet ($125.75'' / 12'' = 10.479 \text{ ft.} \rightarrow 10.5 \text{ ft.}$)

8) Using the chart, $10' 5'' = 454,325.00$ Gallons

INCHES	GALLONS
9'10"	428,882.80
9'11"	432,517.40
10'0"	436,152.00
10'1"	439,786.00
10'2"	443,421.20
10'3"	447,055.80
10'4"	450,690.40
10'5"	454,325.00
10'6"	457,959.60
10'7"	461,591.20
10'8"	465,228.80
10'9"	468,863.10
10'10"	472,498.00
10'11"	476,132.60
11'0"	479,767.20
11'1"	483,401.80

9) Compare the amount above with the amount on the printed ticket. (Printing instructions in section 1.3.1 below.

1.3 PREPARING FOR DELIVERY

1) Ensure the fuel containment tarp is deployed correctly, no rips or tares, etc.



- 2) Verify the fuel truck is positioned correctly on the containment area where the sides can lift containing fuel if a leak occurs.
- 3) Print two tickets from the printer located on the side of bldg. 58 before adding fuel to the tank.

1.3.1 PRINTER INSTRUCTIONS

1. Locate the printer.



2. Open the weatherproof door and press the print button (once per ticket)



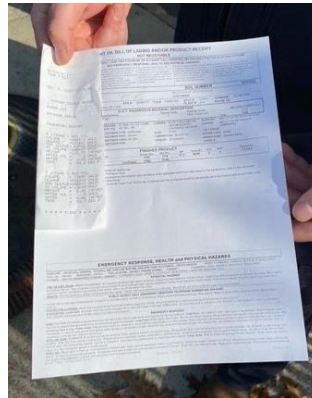
3. Close the weatherproof door.
- 4) Keep one copy for plant records and give one copy to the driver.
- 5) Open the gate closest to the tank to be filled so the driver can access the fill port.

Please see tank 1 & 2 fill port locations below



I.4 DELIVERY COMPLETION

- 1) After all fuel has been added to the tank, print two more tickets. (Printing instructions in section 1.3.1)
- 2) As before, keep one ticket for plant records and give one to the driver for their records.
- 3) Ensure you receive a Bill of Lading from the driver.



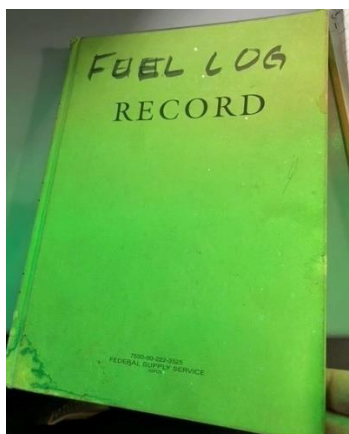
- 4) Verify the covers are placed back over the fill access.



- 5) Secure the containment fence before returning to the CUP.

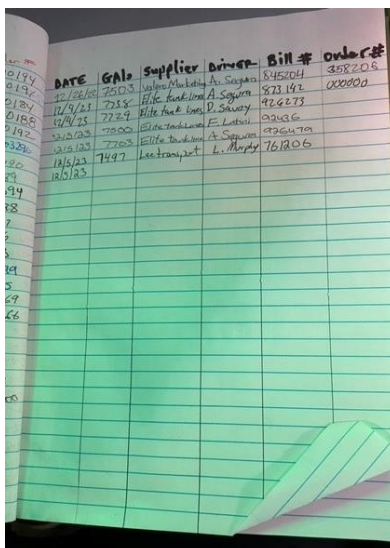
1.5 RECORD FUEL DELIVERY INFO

6) Locate the Fuel Logbook at the log desk.



7) Using the Bill of Lading, Fill in each field listed in the fuel logbook.

- Date
- Gallons
- Supplier
- Driver
- Bill #
- Order #



DATE	Gals	Supplier	Driver	Bill #	Order #
12/26/19	250.3	Valley Marketing	Dr. Simpson	845204	358208
12/27/19	721.8	Valley Marketing	A. Simpson	873142	000000
12/28/19	722.4	Valley Marketing	D. Simpson	873142	000000
12/29/19	722.4	Valley Marketing	E. Simpson	873142	000000
12/30/19	722.4	Valley Marketing	F. Simpson	873142	000000
12/31/19	722.4	Valley Marketing	G. Simpson	873142	000000
1/1/20	722.4	Valley Marketing	H. Simpson	873142	000000
1/2/20	722.4	Valley Marketing	I. Simpson	873142	000000
1/3/20	722.4	Valley Marketing	J. Simpson	873142	000000
1/4/20	722.4	Valley Marketing	K. Simpson	873142	000000
1/5/20	722.4	Valley Marketing	L. Simpson	873142	000000

8) After filling out the information above, staple the tickets and Bill of Lading together and place in the basket located on Dax Sadler's office door.



2. MAIN STORAGE TANK CAPACITY CHART

MAIN STORAGE TANK (Gallons / Capacity)							
INCHES	GALLONS	INCHES	GALLONS	INCHES	GALLONS	INCHES	GALLONS
1"	3,634.60	3'4"	145,384.00	6'7"	287,133.40	9'10"	428,882.80
2"	7,269.20	3'5"	149,018.60	6'8"	290,768.00	9'11"	432,517.40
3"	10,903.80	3'6"	152,653.20	6'9"	294,402.60	10'0"	436,152.00
4"	14,538.40	3'7"	156,287.80	6'10"	298,037.20	10'1"	439,786.00
5"	18,173.00	3'8"	159,922.40	6'11"	301,671.80	10'2"	443,421.20
6"	21,807.60	3'9"	163,557.00	7'0"	305,306.40	10'3"	447,055.80
7"	25,442.20	3'10"	167,191.60	7'1"	308,941.00	10'4"	450,690.40
8"	29,076.20	3'11"	170,826.20	7'2"	312,575.60	10'5"	454,325.00
9"	32,711.40	4'0"	174,460.80	7'3"	316,210.30	10'6"	457,959.60
10"	36,346.00	4'1"	178,095.40	7'4"	319,844.80	10'7"	461,591.20
11"	39,980.60	4'2"	181,730.00	7'5"	323,479.40	10'8"	465,228.80
1' 0"	43,615.20	4'3"	185,364.60	7'6"	327,114.00	10'9"	468,863.10
1'1"	47,249.80	4'4"	188,999.20	7'7"	330,748.60	10'10"	472,498.00
1'2"	50,884.40	4'5"	192,633.80	7'8"	334,383.20	10'11"	476,132.60
1'3"	54,519.00	4'6"	196,268.40	7'9"	338,017.80	11'0"	479,767.20
1'4"	58,153.60	4'7"	199,903.00	7'10"	341,652.40	11'1"	483,401.80
1'5"	61,788.20	4'8"	203,537.60	7'11"	345,287.00	11'2"	487,036.40
1'6"	65,442.80	4'9"	207,172.20	8'0"	348,921.60	11'3"	490,671.00
1'7"	69,057.40	4'10"	210,806.00	8'1"	352,556.20	11'4"	494,305.50
1'8"	72,692.00	4'11"	214,441.00	8'2"	356,190.80	11'5"	497,940.20
1'9"	76,326.60	5'0"	218,076.00	8'3"	359,825.40	11'6"	501,574.80
1'10"	79,961.20	5'1"	221,710.60	8'4"	363,460.00	11'7"	505,209.40
1'11"	83,595.80	5'2"	225,345.20	8'5"	367,094.60	11'8"	508,844.00
2'0"	87,230.40	5'3"	228,979.80	8'6"	370,729.20	11'9"	512,478.60
2'1"	90,865.50	5'4"	232,614.40	8'7"	374,363.80	11'10"	516,113.20
2'2"	94,499.60	5'5"	236,249.00	8'8"	377,998.40	11'11"	519,747.80
2'3"	98,213.40	5'6"	239,888.60	8'9"	381,633.00	12'0"	523,382.40
2'4"	101,768.80	5'7"	243,518.20	8'10"	385,267.60	12'1"	527,017.00
2'5"	105,403.40	5'8"	247,152.80	8'11"	388,902.20	12'2"	530,651.60
2'6"	109,038.00	5'9"	250,787.40	9'0"	392,536.80	12'3"	534,286.20
2'7"	112,672.60	5'10"	254,422.00	9'1"	396,171.40	12'4"	537,920.80
2'8"	116,307.20	5'11"	258,056.60	9'2"	399,806.00	12'5"	541,555.40
2'9"	119,941.80	6'0"	261,691.20	9'3"	403,440.60	12'6"	545,190.80
2'10"	123,576.40	6'1"	265,325.80	9'4"	407,075.20	12'7"	548,824.60
2'11"	127,211.00	6'2"	268,960.40	9'5"	410,709.80	12'8"	552,459.20
3'0"	130,845.60	6'3"	272,596.00	9'6"	414,344.40	12'9"	556,098.80
3'1"	134,480.20	6'4"	276,229.00	9'7"	417,979.00	12'10"	559,728.40
3'2"	138,114.80	6'5"	279,864.20	9'8"	421,613.60	12'11"	563,363.00
3'3"	141,749.40	6'6"	283,498.80	9'9"	425,248.20	13'0"	566,977.60



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4. **REVISION HISTORY**

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Approvals

Chief Operations and Maintenance Section _____

Deputy Chief Utilities Generation Branch _____

Chief Utilities Generation Branch _____